

Sumaira Khan

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EDUCATION

Habib University

BSc Electrical Engineering

Karachi, Pakistan

Aug 2020 - May 2024

Purdue University

Summer Research Program

West Lafayette, USA

May 2023 - July 2023

HONORS AND AWARDS

Research Abroad Scholar: Selected from a pool of thousands of applicants across Pakistan for a fully funded summer research program funded by USEFP for the Summer Undergraduate Research Fellowship (SURF) May 2023 - July 2023

First Runner Up at IFTP: 2nd Position out of 35 universities across the globe in Invent For The Planet 2023 at Texas A&M University, USA April 2023

HU TOPS Scholar: 100% Merit Based Scholarship awarded to cover tuition fee for 4 years. Aug 2020

EXPERIENCE

Research Assistant, Habib University

Dec 2025 - Present

- Courses taught: EE/CE 366L Introduction to Robotics Lab, EE/CE 376L/332L Microcontrollers & Interfacing Lab, EE/CE 100 Intro to IECE.
- Redesigned senior capstone evaluation framework, grading and stage-gate milestones
- Mentored 7 teams (35 students) for a national robotics competition

Assistant Manager - R&D Projects, House of Habib

August 2024 - Dec 2025

- Conducting research on LFP, NCA, and NMC battery technologies, including literature review, use-case analysis, and compliance assessment across IEC/IEEE standards, and international EV/BESS regulatory frameworks.
- Core member of the Business Innovation & Development department, leading multiple cross-functional technical projects
- Developing a locally manufactured transformer oil-level sensor and integrated health-monitoring system.
- Assessed PCB stator motors (90% efficient) for the Pakistani pump market in collaboration with ECM (USA). Conducted hands-on testing on 20+ pumps to evaluate performance gaps and potential for improvement.

Instrumentation and Control Intern, Engro Powergen Thar Ltd. (EPTL)

July 2024 - Aug 2024

- Worked as a Instrumentation and Control Intern at Engro's 2x330MW Coal fired power-plant
- Developed an understanding of a wide range of instruments used in the plant. Observed the plant's Distributed Control Systems (DCS) and various Programmable Logic Controllers (PLCs), and took the opportunity to study their usage and functionality

Summer Undergraduate Researcher, Purdue University

May 2023 - Aug 2023

- Worked on the "Origami-Inspired Multi-Type Locomotive Planetary Exploration Robot." The project involved programming and constructing an autonomous multi-modal locomotive origami robot for space exploration.

PROJECTS

Smart Energy Management Application for Renewable Energy Penetration: For my final year project, I developed a smart Python-based tool for integrating distributed energy resources (DERs) into distribution networks, enabling real-time data processing, automated modeling, visualization to support voltage control and hosting capacity analysis, applied optimization techniques to minimize losses, maintaining voltage stability across the grid. (May 2024)
[IEEE Publication](#)

Load Flow Analysis Considering Voltage Control Modes of Smart Inverters Developed an enhanced load flow analysis algorithm to model smart inverter behavior (Volt-VAR and Volt-WATT modes) in distribution networks, enabling more accurate analysis of renewable-driven grid fluctuations and inverter-based regulation. (April 2024)

Robotic Pick and Place System with Vision-Guided Control: Developed a vision-guided automation pipeline using PhantomX arm with object detection and optimization algorithms.(May 2023)

Two-Wheeled Self-Balancing Robot using PID Controller: Built a two-wheeled self-balancing robot using PID control for real-time stability; demonstrated strong skills in control systems and embedded programming. (Dec 2022)

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

General Secretary & Treasurer IEEE Student Chapter

Jun 2022 - Apr 2024

- Organized multiple workshops and seminars to promote technical knowledge among the student body

Student Mentor for Electrical Engineering batch of 2025

Aug 2021 - Dec 2021

- Worked as a mentor for the ECE batch of 2025 and introduced them to their technical courses

General Secretary, Women in Science and Engineering (WiSE)

Jul 2021 - Feb 2022

- Served as the WiSE Habib University's General Secretary. I helped run the club and organize local events and talks related to ECE and Graduate school.

SKILLS SUMMARY

Python	C++	MATLAB	LaTeX	Ardunio	Raspberry Pi
OpenDSS	PSSE	Orcad PSpice	Simulink	Verilog	Ubuntu
FPGA	TivaC Launchpad	STM32	Ubidots	Design Thinking	Jira